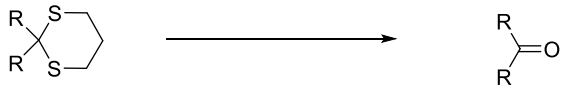
Class 9_CH2204_Sample Solution

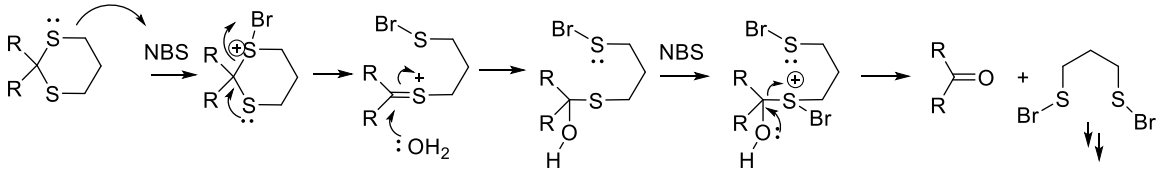
1



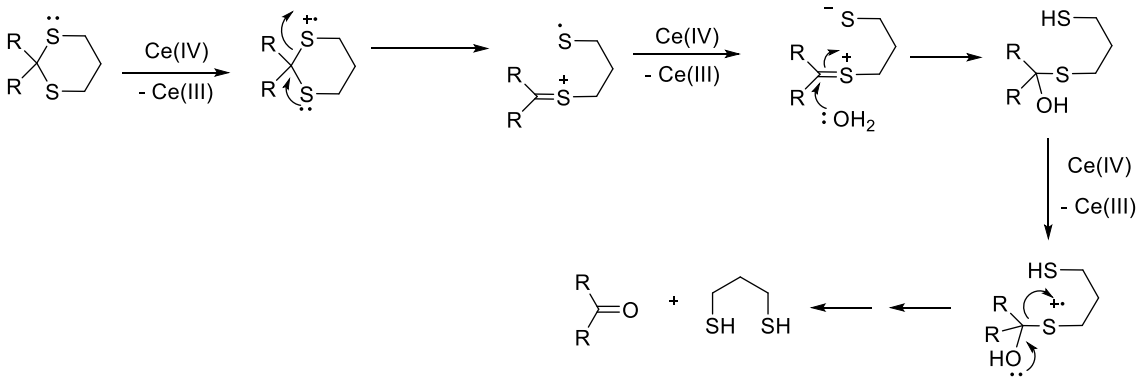
i)NBS,H_2O/MeCN
ii)$Ce(NH_4)_2(NO_3)_6$,H_2O/MeCN
iii)MeI,H_2O/MeOH
iv)$HgCl_2$/H_2O

Solutions

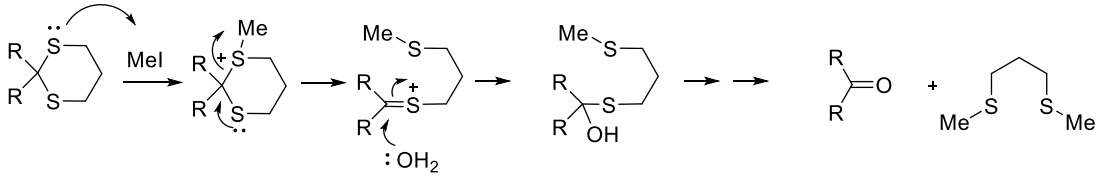
i) NBS acts as electrophilic Bromine source ($Br^{\oplus}$) & the Sulphur atom of thioketal acts as nucleophile to trap $Br^{\oplus}$



ii) Ce(IV) acts as one-electron oxidant to oxidize the sulphur atom of the thioketal



iii)



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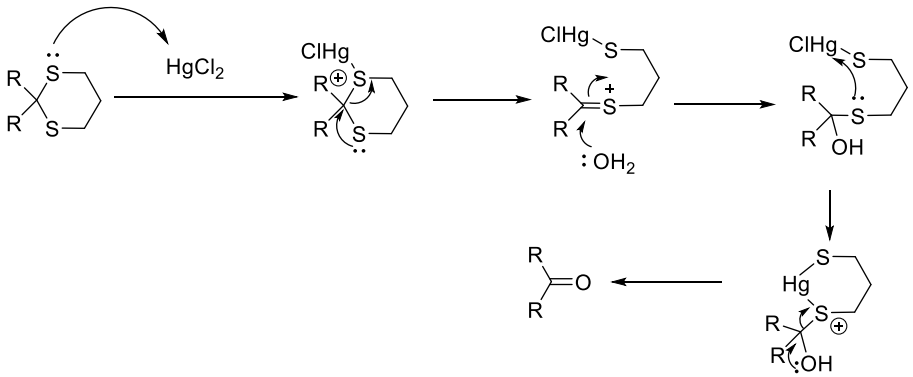

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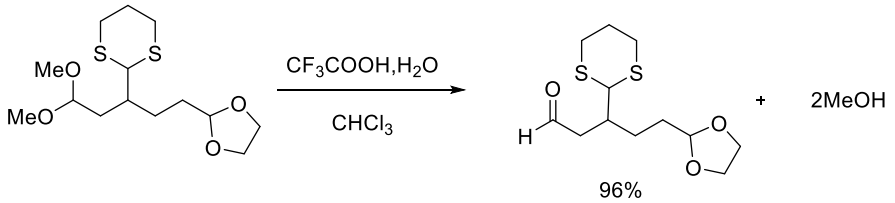
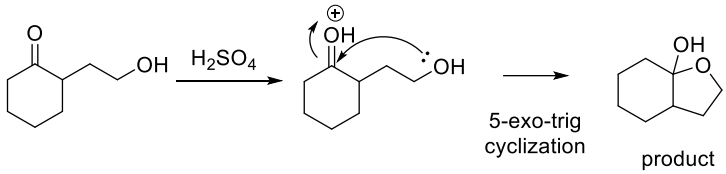
2)



If the given reagents perform acetal deprotection with acyclic one it will give product P and 2 molecules of MeOH

3)



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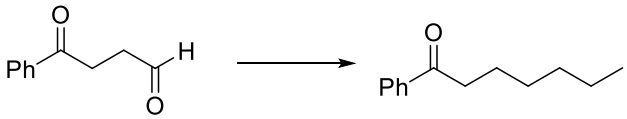
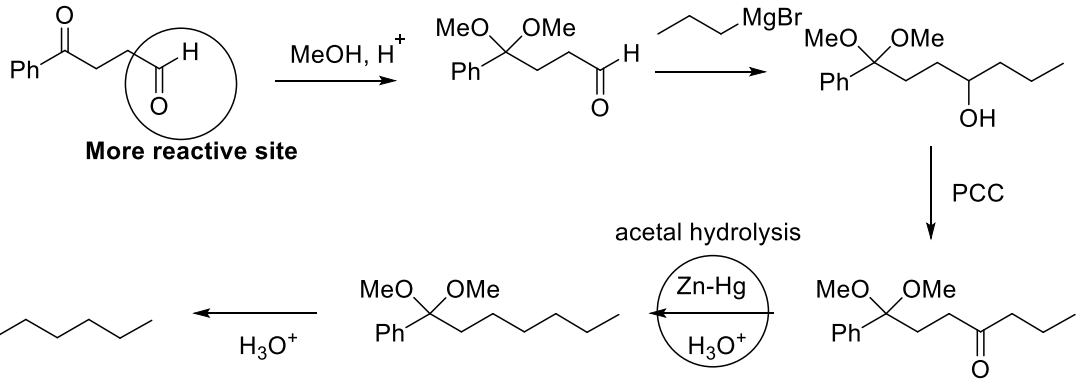
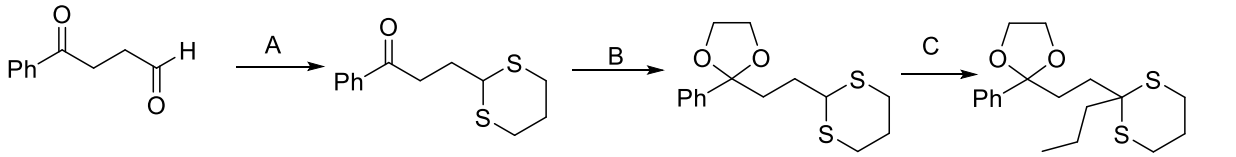


Solutions



proposed forward synthesis



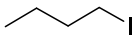
A. =

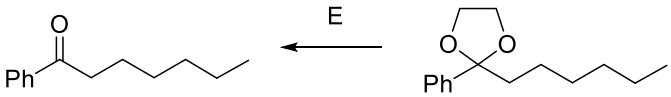
TsOH cat, PhMe, Δ

B =

TsOH cat, PhMe, Δ

D

C = ^{n}BuLi, 



D = H_2/Raney Ni
(Mozingo reduction)

E = H_3O^+

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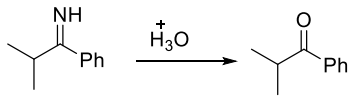
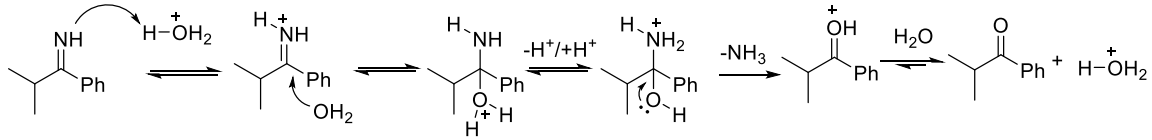
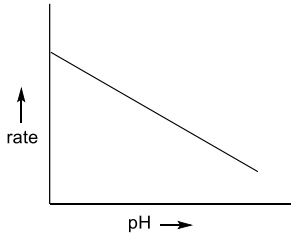
5)



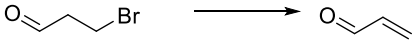
Solution



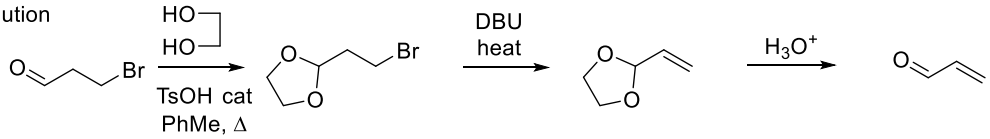


6)

i)



Solution



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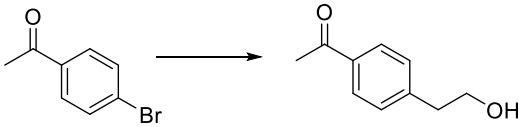
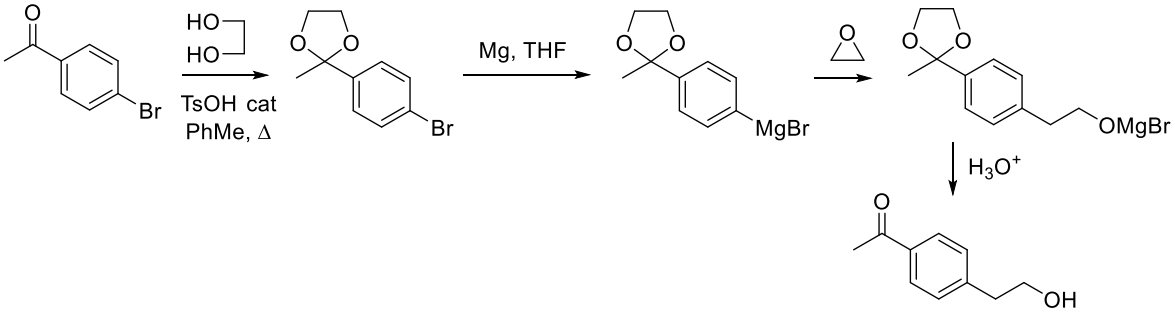
Solution